Chapter-6

Continuous Distribution

The exponential Distribution

Q. what do you mean by exponential dist. Find its application and mean also.

The exponential distribution has density function

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$
$$= 0, \quad \text{otherwise}$$

where λ is real, positive constant.

Application -

1) Measure expected time for an event to occur

Mean
$$E = \int_0^\infty x \lambda e^{-\lambda x} dx$$
$$= \lambda \left[\frac{x e^{-\lambda x}}{-\lambda} - \int_0^\infty \frac{e^{-\lambda x}}{-\lambda} dx \right]$$
$$= \lambda \left[\frac{x e^{-\lambda x}}{-\lambda} + \frac{1}{\lambda} \int_0^\infty e^{-\lambda x} dx \right]$$
$$= \left[-x e^{-\lambda x} - \frac{e^{-\lambda x}}{\lambda} \right]_0^\infty$$
$$= \left[\frac{x}{e^{\lambda x}} - \frac{1}{\lambda e^{\lambda x}} \right]_0^\infty$$

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$$= \left[-(\infty - 0) - \infty \cdot \infty + \frac{1}{\lambda} \right]$$

$$= \frac{1}{\lambda}$$

variance

$$v(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = \int_0^\infty x^2 \lambda e^{-\lambda x} \, dx$$

$$= \left[-x^2 e^{-\lambda x} \Big|_0^\infty + 2 \int_0^\infty x e^{-\lambda x} \right]$$

$$= 0$$

$$v(x) = \frac{1}{\lambda^2}$$

$$M_x(t) = \left(1 - \frac{t}{\lambda}\right)^{-1}$$

CDF $$F(x) = \int_0^x \lambda e^{-\lambda t} \, dt$$

$$\begin{cases} = 1 - e^{-\lambda x}, & x \geq 0 \\ = 0 & x < 0 \end{cases}$$

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Ex 6.4 $\lambda = 10^{-5}$

bet $P\left(T \leq \frac{1}{\lambda}\right)$

$$= \int_0^{1/\lambda} \lambda e^{-\lambda x} dx$$

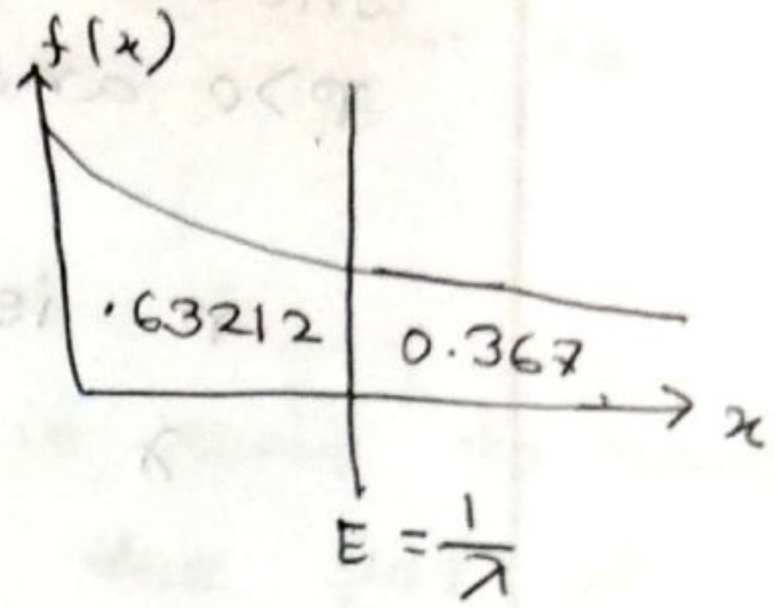
$$= \left[-e^{-\lambda x}\right]_0^{1/\lambda}$$

$$= e^{-1} + e^{0}$$

$$= 1 - e^{-1}$$

$$= 0.63212$$

$$= 63.212\%$$



Gamma Distribution

$$\Gamma n = \int_0^{\infty} x^{n-1} e^{-x} dx$$

$$\Gamma \frac{1}{2} = \int_0^{\infty} x^{\frac{1}{2}-1} e^{-x} dx = \sqrt{x}$$

$$\Gamma n = (n-1)\,\Gamma n-1$$

$$\Gamma n = (n-1)!$$

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Q. What is gamma distribution and its property?

~~for~~ ~~$x>0$~~,

A for $n>0$ the gamma function is

$$\sqrt{n} = \int_0^\infty x^{n-1} e^{-x} dx$$

A continuous variable X is said to have a gamma distribution if the pdf of X is

~~$f(x) = \int$~~

$$f(x) = \frac{\lambda}{\sqrt{r}} (\lambda x)^{r-1} e^{-\lambda x}, \quad x>0$$

where parameter $r>0$, $\lambda>0$

parameter r is called the shape parameter

" λ " " " scale "

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The weibull Distribution

the random variable x is said to have a weibull distribution if the density function is -

$$f(x) = \frac{\beta}{\delta}\left(\frac{x-\gamma}{\delta}\right)^{\beta-1} \exp\left(-\left(\frac{x-\gamma}{\delta}\right)^{\beta}\right)$$

$$= 0 \quad \text{otherwise.}$$

Its parameters are $\gamma\ (-\infty < \gamma < \infty)$ is the location parameter

~~sela~~ scale parameter $\delta > 0$

shape " $\beta > 0$

Application -

as a model for time to failure in electrical and mechanical components and systems.

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Mean $E(X) = \gamma + \delta\Gamma\left(1+\frac{1}{\beta}\right)$

$$V(X) = \delta^2\left\{\Gamma\left(1+\frac{2}{\beta}\right) - \left[\Gamma\left(1+\frac{1}{\beta}\right)\right]^2\right\}$$

$$F(X) = 1 - \exp\left[-\left(\frac{x-\gamma}{\delta}\right)^{\beta}\right] \quad x \geq \gamma$$

the Weibull CDF $F(x)$ is conveniently provided by software packages such as Excel and Minitlab.

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Tama shil

ID: 342

Exercise 6.13

Here $E = 3$

$\therefore \frac{1}{\lambda} = 3$

The percentage of policies will have to pay a claim,

$$P(x<1) = \int_0^1 \lambda e^{-\lambda x}\, dx$$

$$= \left[-\lambda e \frac{-\lambda x}{\lambda} \right]_0^1$$

$$= 1 - e^{-\lambda x}$$

$$= 1 - e^{\frac{-1}{3}}$$

$$= 0.283$$

$$= 28.3\%$$

$\therefore$ 28.3% policies result in a claim.

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Exercise 6.19:

A random variable X, said to have exponential distribution function if it has the density function

$$f(x) = \lambda e^{-\lambda x} \quad ; \quad x > 0$$
$$= 0 \qquad \text{otherwise.}$$

Now $P(X \leq 2) = \int_0^2 \lambda e^{-\lambda x}\, dx$

$$= \left[-e^{-\lambda x}\right]_0^2$$
$$= 1 - e^{-2\lambda}$$

$$P(x \leq 3) = \int_0^3 \lambda e^{-\lambda x}\, dx$$
$$= 1 - e^{-3\lambda}$$

Given, $P(X \leq 2) = \frac{2}{3} P(x \leq 3)$

$$1 - e^{-2\lambda} = \frac{2}{3}\left(1 - e^{-3\lambda}\right)$$

$$\Rightarrow \quad 3 - 3e^{-2\lambda} = 2 - 2e^{-3\lambda}$$

$$\Rightarrow \quad 2e^{-3\lambda} - 3e^{-2\lambda} + 1 = 0$$

$$\Rightarrow \quad 2(e^{-\lambda})^3 - 3(e^{-\lambda})^2 + 1 = 0$$

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let $Y = e^{-\lambda}$

So, $2Y^3 - 3Y^2 + 1 = 0$

$(Y-1)^2 (2Y+1) = 0$

Here $Y = 1, -\frac{1}{2}$

$e^{-\lambda} = 1$ $\quad$ $e^{-\lambda} = -\frac{1}{2}$

$\lambda = 0$

Only $\lambda = 0$ satisfies this condition; but we must have $\lambda > 0$.

So there is no value for which

$$P(X \leq 2) = \frac{2}{3} P(X \leq 3)$$

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Exercise 6.15:

Given,

$$C_I = C; \quad x > 15$$

$$= C + Z; \quad x \leq 15$$

$$\lambda_I = \frac{1}{25}$$

$$C_{II} = 3C; \quad x > 15$$

$$= 3C + Z; \quad x \leq 15$$

$$\lambda_{II} = 35$$

Now, $E(C_I) = CP(X > 15) + (C+Z)\, P(X \leq 15)$

$$= C \int_{15}^{\infty} \lambda_I e^{-\lambda_I x}\, dx + (C+Z) \int_{0}^{15} \lambda_I e^{-\lambda_I x}\, dx$$

$$= C \left[-e^{\frac{-x}{25}}\right]_{15}^{\infty} + (C+Z) \left[-e^{-\frac{x}{25}}\right]_{0}^{15}$$

$$= Ce^{\frac{-15}{25}} + (C+Z)\left(1 - e^{\frac{-15}{35}}\right)$$

$$= C + Z\, 0.4512Z$$

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$$E(C_{II}) = 3C\,P(X>15) + (3C+Z)P(X\leq 15)$$

$$= 3C\,e^{\frac{-15}{35}} + (3C+Z)\left(1 - e^{\frac{-15}{35}}\right)$$

$$= 3C + 0.3486Z$$

$$E(C_{II}) - E(C_I) = 2C - 0.1026Z$$

So we can recommend process I if

$C > 0.0513Z$.

Exercise 6.16:

$$E = 20000$$

$$\frac{1}{\lambda} = 20000$$

$$\lambda = \frac{1}{20000}$$

~~time~~

Now $P(\cancel{x} > 10000$

the probability that transistore fails by 30,000 hours

$$P(X > 30000 \mid X > 20000)$$

$$= P(X > 10000)$$

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$$= \int_{10000}^{\infty} \lambda e^{-\lambda x} dx$$

$$= \left[-e^{-\lambda x} \right]_{10000}^{\infty}$$

$$= e^{\frac{-10000}{20000}}$$

$$= 0.6069.$$

$$\therefore \quad P(X < 30000 \mid X > 20000)$$

$$= 1 - 0.6069$$

$$= 0.3936.$$

— o —

Ex 6.34:

Here location parameter $\gamma = 0$

scale " $\delta = 1$

shape parameter $\beta = 1, 2, 3, 4, 5, 6, 7$

the weibull distribution is

$$f(x) = \frac{\beta}{\delta} \left(\frac{x-\gamma}{\delta} \right)^{\beta-1} \exp \left[-\left(\frac{x-\gamma}{\delta} \right)^{\beta} \right] \quad x \geq \gamma$$

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$$f(x) = \frac{\lambda}{\Gamma r} (\lambda x)^{r-1} e^{-\lambda x} \quad x > 0$$

$$\Gamma = 0! = 1$$

where

$r > 0$ and $\lambda > 0$.

r is called shape parameter

λ " " scale "

if $r = 1$

$$f(x) = \lambda (\lambda x)^{0} e^{-\lambda x} \quad ; x > 0$$

$$= \lambda e^{-\lambda x} \quad ; x > 0$$

if $x_j \sim Exp(\lambda)$

$$f(x_j) = \lambda e^{-\lambda x_j} \quad x_j > 0$$

$$E(x) = \frac{r}{\lambda} = \sum_{j=1}^{r} E(x_j)$$

$$V(x) = \frac{r}{\lambda^2}$$

$$M_x(t) = \left(1 - \frac{t}{\lambda}\right)^{-r}$$

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for $\gamma = 0, \quad \delta = 1$

$$f(x) = \beta (x)^{\beta-1} \exp\left(-x^{\beta}\right)$$

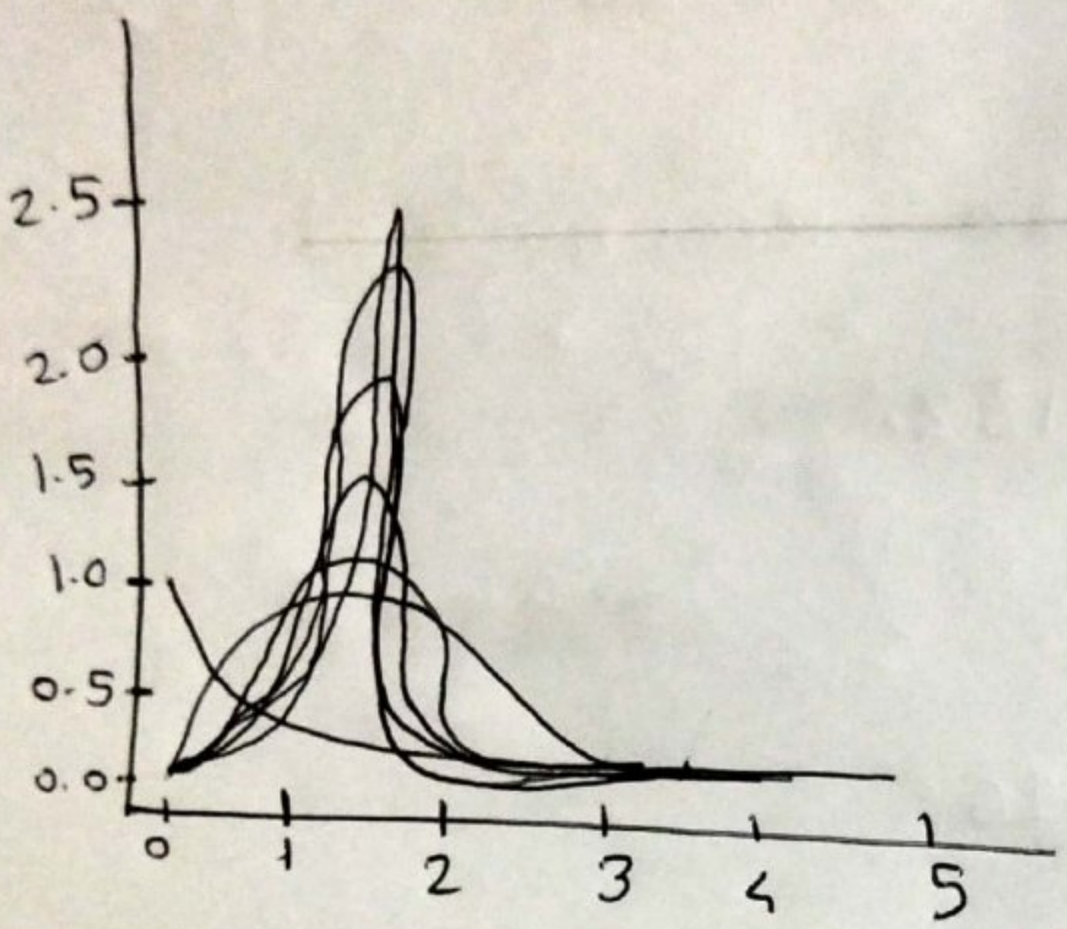


fig: Weibull distribution for $\beta = 1$ to 7.

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Exercise 6.35 :

We know

$$F(x) = 1 - e^{-\left(\frac{x-\gamma}{\delta}\right)^{\beta}} \qquad x \geq \gamma$$

here, $\gamma = 0$, $\beta = \frac{1}{4}$, & $\delta = 200$

$$\therefore F(x) = 1 - e^{-\left(\frac{x-0}{200}\right)^{1/4}}$$

a) the fraction of these units will survive ~~up~~ to 1000 hours

$$= 1 - F(1000)$$

$$= 1 - 1 + e^{-\left(\frac{1000}{200}\right)^{1/4}}$$

$$= 1 + e^{-5^{1/4}}$$

$$= 1 + e^{-1.49}$$

$$= e^{-1.49}$$

$$= 224$$

b) $E(X) = \gamma + \delta \Gamma\left(\frac{1}{\beta} + 1\right)$

$$= 0 + 200\, \Gamma(5)$$

$$= 200\,(4 \times 3 \times 2 \times 1)$$

$$= 4800.$$

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